EASY · ALGEBRA · ANSWER KEY

Linear functions

07

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

D

D. The estimated height of the tree was 4 feet when it was first measured.

Choice D is correct. It's given that the function $f x = 8x + 4$ gives the estimated height, in feet, of a willow tree x years after its height was first measured. For a function defined by an equation of the form $f x = mx + b$, where m and b are constants, b represents the value of $f x$ when $x = 0$. It follows that in the given function, 4 represents the value of $f x$ when $x = 0$. Therefore, the best interpretation of 4 in this context is that the estimated height of the tree was 4 feet when it was first measured.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q2

11

The correct answer is 11. It's given that the function $f x = 14 + 4x$ represents the total cost, in dollars, of attending an arcade when x games are played. Substituting 58 for $f x$ in the given equation yields $58 = 14 + 4x$. Subtracting 14 from each side of this equation yields $44 = 4x$. Dividing each side of this equation by 4 yields $11 = x$. Therefore, 11 games can be played for a total cost of $\$58$.

Q3 D

D. $f(x) = 2x - 5$

Choice D is correct. An equation defining the linear function f can be written in the form $fx = mx + b$, where m is the slope and $0, b$ is the y-intercept of the graph of $y = fx$ in the xy-plane. It's given that the graph of $y = fx$ has a slope of 2. Therefore, $m = 2$. It's also given that the graph of $y = fx$ has a y-intercept at $0, -5$. Therefore, $b = -5$. Substituting 2 for m and -5 for b in the equation $fx = mx + b$ yields $fx = 2x - 5$. Thus, the equation that defines f is $fx = 2x - 5$.

Choice A is incorrect. For this function, the graph of $y = fx$ in the xy-plane has a slope of $\frac{1}{2}$, not 2.

Choice B is incorrect. For this function, the graph of $y = fx$ in the xy-plane has a slope of $-\frac{1}{2}$, not 2.

Choice C is incorrect. For this function, the graph of $y = fx$ in the xy-plane has a slope of -2, not 2.

Q4 C

C. $c = 11d + 10$

Choice C is correct. It's given that the cost of renting a tent is $\$11$ per day for d days. Multiplying the rental cost by the number of days yields $\$11d$, which represents the cost of renting the tent for d days before the insurance is added. Adding the onetime insurance fee of $\$10$ to the rental cost of $\$11d$ gives the total cost c, in dollars, which can be represented by the equation $c = 11d + 10$.

Choice A is incorrect. This equation represents the total cost to rent the tent if the insurance fee was charged every day.

Choice B is incorrect. This equation represents the total cost to rent the tent if the daily fee was $\$d + 11$ for 10 days.

Choice D is incorrect. This equation represents the total cost to rent the tent if the daily fee was $\$10$ and the onetime fee was $\$11$.

Q5

D

D. The estimated length of the vine plant was 9 inches when Tavon purchased it.

Choice D is correct. It's given that the function f defined by $ft = 14t + 9$ gives the estimated length, in inches, of a vine plant t months after Tavon purchased it. For a function defined by an equation of the form $ft = mt + b$, where m and b are constants, b represents the value of $f0$, or the value of ft when the value of t is 0. Therefore, for the function defined by $ft = 14t + 9$, 9 represents the value of ft when the value of t is 0. This means that 0 months after the vine plant was purchased, the estimated length of the vine plant was 9 inches. Therefore, the best interpretation of 9 in this context is the estimated length of the vine plant was 9 inches when Tavon purchased it.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. The vine plant is expected to grow 14 inches, not 9 inches, each month.

Choice C is incorrect and may result from conceptual or calculation errors.

Q6

B

B. 9

Choice B is correct. Substituting 72 for fx in the given function yields $72 = 8x$. Dividing each side of this equation by 8 yields $9 = x$. Therefore, $fx = 72$ when the value of x is 9.

Choice A is incorrect. This is the value of x for which $fx = 64$, not $fx = 72$.

Choice C is incorrect. This is the value of x for which $fx = 512$, not $fx = 72$.

Choice D is incorrect. This is the value of x for which $fx = 640$, not $fx = 72$.

Q7

D

D. 8

Choice D is correct. The value of $g(0)$ is found by substituting 0 for x in the function g. This yields $g(0) = -0 + 8$, which can be rewritten as $g(0) = 8$.

Choice A is incorrect and may result from misinterpreting the equation as $g(x) = x + (-8)$ instead of $g(x) = -x + 8$. Choice B is incorrect. This is the value of x, not $g(x)$. Choice C is incorrect and may result from calculation errors.

Q8

B

B. $f(x) = 3x - 8$

Choice B is correct. An equation defining a linear function can be written in the form $fx = mx + b$, where m and b are constants, m is the slope of the graph of $y = fx$ in the xy-plane, and $0, b$ is the y-intercept of the graph. It's given that for the function f, the graph of $y = fx$ in the xy-plane has a slope of 3. Therefore, $m = 3$. It's also given that this graph passes through the point $0, -8$. Therefore, the y-intercept of the graph is $0, -8$, and it follows that $b = -8$. Substituting 3 for m and -8 for b in the equation $fx = mx + b$ yields $fx = 3x - 8$. Thus, the equation that defines f is $fx = 3x - 8$.

Choice A is incorrect. For this function, the graph of $y = fx$ in the xy-plane passes through the point $0, 0$, not $0, -8$.

Choice C is incorrect. For this function, the graph of $y = fx$ in the xy-plane passes through the point $0, 5$, not $0, -8$.

Choice D is incorrect. For this function, the graph of $y = fx$ in the xy-plane passes through the point $0, 11$, not $0, -8$.

Q9 A

A. $-12, 0$

Choice A is correct. The x-intercept of a graph is the point where the graph intersects the x-axis. The graph of function f, where $y = fx$, intersects the x-axis at $-12, 0$. Therefore, the x-intercept of the graph of f is $-12, 0$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 D

D. 55

Choice D is correct. In the given equation, s is the speed, in miles per hour, of a certain car t seconds after it began to accelerate. Therefore, the speed of the car, in miles per hour, 5 seconds after it began to accelerate can be found by substituting 5 for t in the given equation, which yields $s = 40 + 35$, or $s = 55$. Thus, the speed of the car 5 seconds after it began to accelerate is 55 miles per hour.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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